Numerical Methods for ODEs: Exercise Week 3

Problem 1:

By using Theorems from the lecture, show that the initial value problem

$$y'(t) = \lambda y(t),\ y(0) = y_0 \tag{1}$$

with $\lambda, y_0 \in \mathbb{R}$, has a unique solution that depends continuously on λ and y_0.

Problem 2:

Let $c \in (0,1)$, e.g. $c = \frac{1}{2}$ or $c = \frac{1}{\pi}$. Show that $f(y) = y^c$ is not Lipshitz continuous in $[0, +\infty)$ (but note that it is continuous).

Problem 3:

Consider again $f(y) = y^c$, for fixed $c \in (0,1)$, together with an initial condition $y_0 = 0$ and the resulting IVP

$$y'(t) = f(y),\ y(0) = y_0. \tag{2}$$

1. Find the analytical solution(s) to the IVP.
2. Create a script in Python to solve the IVP by using `odeint` or `solve_ivp` and plot the numerical result for different constants c, e.g. $c = \frac{1}{2}$ or $c = \frac{1}{\pi}$. For comparison, plot also your analytical result obtained in 1.
3. Change the initial condition to a value ϵ close to zero, say $\epsilon = 10^{-5}$, and plot the numerical solutions again.
4. Describe the plots and explain what is happening.